

```

public static int task10CP(int[] input, int target) {
    int left = 0; // Initialize the left pointer
    int right = input.length - 1; // Initialize the right pointer
    while (left <= right) {
        int mid = left + (right - left) / 2; // Calculate the mid index of the array (left + right) / 2
        if (input[mid] == target) {
            return mid; // If the target element is found, return the target index
        } else if (input[mid] < target) {
            left = mid + 1; // If the target is greater, the target is in the right half
        } else {
            right = mid - 1; // If the target is smaller, the target is in the left half
        }
    }
    return -1; // If the target element is not found, return -1
}

```

```

public static void main(String[] args) {
    int[] input = {2, 5, 8, 12, 16, 23, 38, 45, 56, 72};
    int target = 45;
    int result = task10CP(input, target);
    System.out.println(result);
}

```